CONTACT

monishnareshkumar@gmail.com

+65 8179 3493

Singapore

github.com/Brokenx25/
MONISH-PROJECTS

linkedin.com/in/
monish-naresh-kumar

DETAILS

NATIONALITY
Indian

WORK ELIGIBILITY
Student Pass — studying in Singapore

SKILLS

Python — Advanced
C / C++ / C# — Intermediate
SQL — Intermediate
Lua — Intermediate
HTML / CSS — Intermediate
Git — Advanced

ROS 2 · Webots · CoppeliaSim

OpenCV · MediaPipe · LLMs · 3D printing

LANGUAGES

English — Advanced (C1)
Tamil — Upper Int. (B2)
Hindi — Intermediate (B1)

Monish Naresh Kumar

ROBOTICS & AI ENGINEERING STUDENT

ABOUT ME

Robotics & AI engineering student (B.Eng., graduating September 2026) with a broad hands-on portfolio spanning perception, autonomous navigation, and human–robot interaction. Builds robots and intelligent systems end to end — from hardware assembly and control logic through computer-vision and speech pipelines — primarily in Python and C++ on ROS 2. Quick to learn new tools and effective in teams.

PROJECTS

PERCEPTION & MACHINE LEARNING

Computer Vision Pipeline *Python · OpenCV · MediaPipe*
- Built an end-to-end vision pipeline detecting faces, hand gestures, and body pose, integrating detection, tracking, and classification into one real-time system.

Speech Recognition Pipeline *Python · speech & audio processing*
- Developed a voice pipeline performing both speaker identification and speech-to-text, distinguishing who is speaking from what is being said.

AUTONOMOUS SYSTEMS & NAVIGATION

Autonomous Path-Following Car *Python · path planning · sensor feedback*
- Programmed a self-driving car to follow a planned route while sensing its surroundings and reacting in real time to obstacles, changes, and hazards.

Line-Following Robot — Built from Scratch *C/C++ · microcontroller · IR sensors · 3D printing*
- Designed and 3D-printed the chassis, assembled the wiring and control logic, and tuned the control loop for reliable autonomous line tracking.

Hazard Navigation Simulation *CoppeliaSim · Lua · Python*
- Modelled a real-world minefield scenario to analyse environmental dangers and plan safe navigation routes.

Multi-Behaviour Robot Simulation *Webots · Python*
- Implemented a simulated robot exhibiting 10 distinct autonomous behaviours across sensing, decision-making, and actuation.

HUMAN–ROBOT INTERACTION & SOCIAL ROBOTICS

Medical Robot Dog Assistant *ROS 2 · Python · LLM · computer vision · Final-Year Project*
- Developing an autonomous quadruped on ROS 2 that continuously monitors patients, integrating an LLM for natural interaction and a vision pipeline for perception — aimed as a lower-cost alternative to medical service dogs.

Teaching Assistant Robot *Alpha Mini SDK · Python · Team Project*
- Programmed an Alpha Mini humanoid, in a team of 4, to act as an interactive teaching assistant guiding students through learning difficulties.

Synchronized Robot Performance *Alpha Mini · Python · Web (HTML/CSS) · Independent*
- Coordinated multiple Alpha Mini robots to perform in perfect sync, triggered on demand via a QR-code-accessible web app.

SOFTWARE & SYSTEMS

Othello (Reversi) Game *Python* · **Wireless Device Communication** *C/C++ · wireless protocols*
- Built a fully playable Othello game (rules, move validation, turn logic) and implemented reliable wireless data exchange between two devices.

EDUCATION

Bachelor of Engineering in Robotics and AI — Expected Sep 2026
PSB Academy, Singapore

VOLUNTEERING

Event Volunteer, PSB Academy — 2026
- Coordinated logistics and gathered participant feedback across campus events and programs to improve future delivery.